

Environmental Regulation and Compliance

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Learning Objectives

After taking this class, you will be able to:

- *understand* the theoretical foundation of conventional environmental regulation
- *acknowledge* the main instruments and practices of conventional environmental regulation
- *illustrate* the achievements and challenges of environmental regulation in China

Theoretical Foundation of Conventional Environmental Regulation

"Command-and-Control" Approach of Regulation

- "Command-and-control" was the starting point for both regulators and regulatory scholars. It is still the most widely used regulatory instruments of environmental protection even today.
- "Command-and-control" regulation is implemented primarily through the ***use of rules and standards*** specified in advance and heavily rely on ***hierarchical authority to enforce*** its implementation.
- It requires polluters to meet specific emission-reduction targets (i.e., outcomes) and often requires the installation and use of specific types of equipment to reduce emissions (i.e., procedures).

Key Elements of "Command-and-Control" Approach I

- (ex ante) Rules and Standards:
  - They usually specify a goal, an outcome, or an action that the regulated should achieve
  - Some major institutional evolution of environmental protection in China
    - Action Plan for Prevention and Control of Air Pollution (2013)
    - Action Plan for Prevention and Control of Water Pollution (2015)
    - Environmental Protection Law (2015)
    - Environmental Protection Tax Law (2018)
    - Regulation on the Administration of Permitting of Pollutant Discharges (2021)
  - A list of environmental standards

Key Elements of "Command-and-Control" Approach II

- (ex post) Inspection:
  - It is to investigate whether the goals, outcomes, or actions that required in the rules are realized or implemented.
  - Two ways to conduct regulatory inspection (McCubbins and Schwartz, 1984):
    - Police patrol: "At its own initiative, *regulators* examines a sample of regulated activities, with the aim of detecting and remedying any violations of legislative goals and, by surveillance, discouraging such violations." (**centralized and direct**)
    - Fire alarm: Regulators "establishes a system of rules, procedures, and informal practices that enable *individual citizens and organized interest groups* to examine administrative decisions, to charge violations, and to seek remedies." (**less centralized and indirect**)





  - **Frequent and thorough inspection is a prerequisite for effective regulation and provides foundation for subsequent enforcement and sanctions.**

Key Elements of "Command-and-Control" Approach III

- (ex post) Sanctions:
  - It imposes (social) costs on the regulated entities if the latter do not meet the requirements specified in the rules/standards.
  - There are various instruments of environmental enforcement in China.
    - regulating the enterprises: Fine; Rectification within a prescribed time; Shutting down
    - Supervising the governments: Admonishment Talk, Demotion

Reflective Question:

According to the characteristics of "command-and-control" approach discussed in the previous slides, do you have an intuitive sense about why "command-and-control" could solve environmental problems and deal with negative externalities?

Firm's Payoff under Different Scenarios

- Benefit of production: B
- Cost of production:C_0
- Cost of pollution treatment:C_1
- Penalty against violation: F
- Probability of detection=frequency of inspection: p

	Firm not inspected by regulator	Firm inspected by regulator
Compliance	$B - C_0 - C_1$	$B - C_0 - C_1$
Non-compliance	$B - C_0$	$B - C_0 - C_1 - \mathrm{F}$

Payoff of noncompliance: $B - C_0 - (C_1 + F) * p$

The Rationale of "Command-and-Control" Regulation

- Underlying assumption: "Economic man"
  - Decisions regarding compliance are based on self-interest.
  - Businesses comply only when the costs of noncompliance outweigh the benefits of noncompliance.

But is it always true ?



- Adding (social) cost to the regulated entities (e.g., polluting firms)→ make penalties high enough and the probability of detection great enough that it becomes economically irrational for facilities to violate environmental requirements.

- Deterrence-based enforcement: focusing on detecting and punishing violations

Practices of Environmental Regulation in China

Procedures of Routine Command-and-Control Regulation

- Who conduct regular environmental inspection and enforcement?
  - Law enforcement unit
- What are the mechanisms of regular environmental inspection and enforcement?
  - random inspections by randomly selected law enforcement officers and requiring the prompt release of results （“双随机，一公开”）
  - quick response to community calls

Central Environmental Protection Inspection (CEPI)



Does Central Environmental Protection Inspection (CEPI) work?

Local AQI, PM2.5, and PM10 level have significantly reduced during inspections. This effect is especially substantial for localities with less desired air quality. (Wang, Liu, and Xiong, 2019)

Cities that experience CEPI do not have significant better air quality compared to their adjacent localities. However, when comparing to the nonadjacent localities, cities being inspected by the central inspection team have significant low level of PM10, SO_2, CO, NO_2, but not PM2.5, O_3 (Liu and Wu, 2019).

Challenges of Environmental Regulation in China

Limited Capacity of Inspection I

- Limited number of inspectors and enforcement officials

Year	# of officials in Guangzhou Environmental Protection Bureau	# of industrial enterprises
2000	250	31,416
2006	220	---
2013	280	65,598

(Liu, van Rooij, and Lo, 2018)

Is this because compliance rate is relatively high so that we do not need a large number of inspectors and enforcers? Probably not.

Limited Capacity of Inspection II

- Challenge in monitoring nonpoint source (NPS) pollution

Pollutants	Year	Industrial pollution	Agricultural pollution	Urban domestic pollution	centralized pollution control and treatment facilities
COD (10,000 ton)	2011	354.8	1186.1	938.8	20.1
	2012	338.5	1153.8	912.8	18.7
	2013	319.5	1125.8	889.8	17.7
	2014	311.3	1102.4	864.4	16.5
	2015	293.5	1068.6	846.9	14.5
	2016	122.8	57.1	473.5	4.6
	2017	91	31.8	483.8	2.3
	2018	81.4	24.5	476.8	1.5
	2019	77.2	18.6	469.9	1.4

(Annual Report of Environmental Statistics, 2015; Annual Report of Ecology and Environmental Statistics 2016-2019)

Limited Capacity of Inspection II (Cont.)

- Challenge in monitoring nonpoint source (NPS) pollution

Pollutants	Year	Industrial pollution	Agricultural pollution	Urban domestic pollution	centralized pollution control and treatment facilities
NH-3	2011	28.1	82.7	147.7	2
	2012	26.4	80.6	144.6	1.9
	2013	24.6	77.9	141.4	1.8
	2014	23.2	75.5	138.1	1.7
	2015	21.7	72.6	134.1	1.5
	2016	6.5	1.3	48.4	0.7
	2017	4.4	0.7	45.4	0.3
	2018	4	0.5	44.7	0.2
	2019	3.5	0.4	42.1	0.3

(Annual Report of Environmental Statistics, 2015; Annual Report of Ecology and Environmental Statistics 2016-2019)

Short-term vs. Long-term Policy Effects

- While central inspections are in progress, emissions of a major pollutant sulfur dioxide (SO2), fall on average by 25-27%.
- However, pollution reverts to prior level after inspectors leave.
- Reversion occurs most rapidly among firms accountable to the central government, while cleanup persists longer among firms accountable to local government.

(Karplus and Wu, 2020)

Campaign-Style Enforcement

- Campaign-style enforcement is a type of policy implementation involving extraordinary mobilization of administrative resources under political sponsorship to achieve a specific policy target within a defined period of time.
- It can be either short term or sustained for a prolonged period of time, they may even be institutionalized as routine enforcement measures at the end of the campaign.
- Campaign-style enforcement achieves its goal through both ***resource sufficiency*** and ***power adequacy***.

(Liu, Lo, Zhan, and Wang, 2015)

The Problems of Blunt Force Regulation I

- Blunt Force Regulation (van der Kamp, 2020)
  - "The state applies crude, **one-size-fit-all restrictions** against regulated entities—even against entities that are complying with the law."
  - "The state authorizes bureaucrats to use **extreme measures**—such as forcibly destroying factories—to ensure that regulatory action produces immediate change."
  - "Between 2010 and 2015, authorities in 269 out 287 cities were ordered to forcibly reduce or halt production in dozens of highly polluting industries."

The Problems of Blunt Force Regulation II

- Reduce the expected benefits of compliance → disincentivize compliant behavior
- Damage the trust between the regulators and the regulated
- Cannot sustain the threat of detection in the long term

Compliance and beyond-compliance

- Command-and-control regulation offers no incentive to improve the quality of the environment beyond the standard set by a particular law. Once the command-and-control regulation has been satisfied, polluters have zero incentive to do better.
- Command-and-control regulation is inflexible. It usually requires the same standard for all polluters, and often the same pollution-control technology as well.

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